

Syed Tahir Ali Shah

Ph.D. Researcher in Mechanical Engineering, Embedded IoT, Affective Computing, and Smart Mobility

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Centre for Mechanical Technology and Automation (TEMA), Department of Mechanical Engineering, University of Aveiro, Portugal

ORCID Google Scholar Ciencia Vitae LinkedIn

Profile

Multidisciplinary Ph.D. researcher working across Mechanical Engineering, Computer Science, embedded Internet-of-Things systems, wearable sensing, edge intelligence, and human-centred smart mobility. Current research focuses on emotion-aware and physiology-aware mobility systems that use wearable signals, embedded machine learning, and IoT integration to support safer and more adaptive mobility experiences.

Education and Research Appointments

Ph.D. in Mechanical Engineering 2023/11/01–2027/06/30 (ongoing)

University of Aveiro, Portugal

Focus: emotion-aware IoT-integrated mobility systems. **Research Trainee** From 2026/01/01

Fundacao para a Ciencia e a Tecnologia, Portugal

Host: University of Aveiro, Department of Mechanical Engineering. **Scholarship Holder** 2023/11/01–2025/12/31

University of Aveiro, Portugal

Ph.D. research work in the AM2R and MariaBike research context. **M.Sc. Computer Science** 2016/01/10–2018/08/31

Iqra University - Islamabad Campus, Pakistan

Classification: 14/20. **B.Sc. Computer Science** 2011/06/01–2016/01/01

Karakoram International University, Pakistan

Major: software engineering, database systems, data structures, and algorithm design.

Research Interests

- Affective computing and emotion-aware IoT systems.
- Embedded machine learning, TinyML, mobile runtimes, and edge intelligence.
- Wearable-driven physiological monitoring for mobility safety.
- Human-centred smart mobility, e-bikes, and intelligent transport systems.

Selected Publications

- Syed Tahir Ali Shah, J.P. Santos, Gabriel Constantinescu, Jose M. Fernandes, A.B. Pereira. “Physio-Digital Twin for Human-Centered IoT Mobility: A Proof-of-Concept Implementation on the MariaBike Platform.” ICICT 2026. Best Paper Award. DOI: 10.1145/3803291.3803295.
- Syed Tahir Ali Shah, J. M. Fernandes, J. P. Santos, G. Constantinescu, Antonio B. Pereira. “The Micro-Mobility Sensing Gap: A Systematic Review of Physiological Safety Monitoring from Cycling to E-Scooters.” Sensors, 2026. DOI: 10.3390/s26041110.

- Gabriel Constantinescu, Syed Tahir Ali Shah, Jose Paulo Oliveira Santos, Joao Manuel Cardoso, Mario Jorge de Sousa Henriques, Antonio Manuel de Bastos Pereira. "Tensile and Flexural Behavior of Bi-axial Non-Crimp-Fabric Composites for Two-Wheeled Electric-Vehicle Chassis." *Fibers*, 2026. DOI: 10.3390/fib14050061.
- Syed Tahir Ali Shah. "Extending IoT Networks with the Human Factor: An Example for an E-bike Scenario." 7th International Conference on Technologies for the Wellbeing and Sustainable Manufacturing Solutions, 2024. DOI: 10.48528/mehv-z227.

Projects and Software

MariaBike Dashboard	2025
<i>Featured software</i>	
Real-time e-bike dashboard with BLE and MQTT integration for physiological monitoring in mobility. FCT	
BDANA PhD Work Programme	2026–2030
<i>Ph.D. studentship</i>	
Embedded intelligent systems for personalized and sustainable mobility. AM2R Mobilizing Agenda	
2025	
<i>Ph.D. student fellow</i>	
Business innovation in the two-wheel sector at the University of Aveiro.	

Service and Mentorship

Scientific Mentorship (CHAI I)	2024/10/01–2025/06/30
<i>University of Aveiro</i>	
Mentoring student Joao Bispo in Scientific Mentorship I. Thesis Supervisor	2022/06/01–2023/06/30
<i>University of Baltistan, Pakistan</i>	
Topic: IoT-Based Smart Farming System with Real-Time Soil Moisture and Weather Monitoring. Thesis	
Co-supervisor	2020/01/01–2021/01/31
<i>University of Baltistan, Pakistan</i>	
Topic: Real-Time Emotion Recognition from Facial Expressions Using Deep Learning.	

Languages

Urdu	Native / mother tongue
English	Upper intermediate to proficient across communication skills
Portuguese	Beginner to intermediate working proficiency